# Transitive Colouring – Dodecahedron

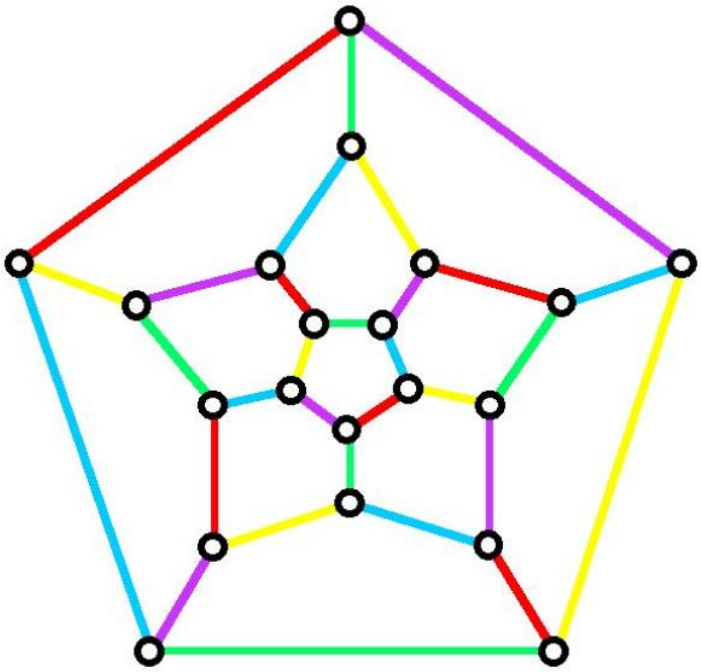


*Figure 1. Transitive colouring of the dodecahedron (credit: Geoffrey Pearce)*

One thing that often gets overlooked in modular origami is the difficult choice of which colours to use, and where to use them. Here is an example of how mathematics, in particular the study of symmetry known as “group theory”, can be exploited to create a more aesthetically pleasing result.

The collection of dots and lines in Figure 1 is known as a “graph” and gives us a 2D representation of the dodecahedron (the solid with 12 pentagonal faces). Think of building a cube out of stretchy wire and then flattening it against a wall. Graphs allows us to create a “map” that we can follow in the assembly process, which we will call a “colouring”.

The idea is to come up with a colouring that is “symmetric” in the sense that if we view the object from any angle, it will look the same (up to the swapping of colours). One way to do this is to use five colours and arrange them in what is known as an “A5-transitive 1-decomposition” as shown in Figure 1. This basically means that we can apply any rotation and the colouring remains the same, and that no two edges of the same colour touch. For more information, see Geoffrey Pearce’s doctoral thesis:

https://research-repository.uwa.edu.au/en/publications/transitive-decompositions-of-graphs/